Mitigating Hallucination in Large Multimodal Models via Efficient Visual Attention Regularization

CISC 874 Project Report: Maxwell Woodfield

***Abstract*—Large Multimodal Models (LMMs) exhibit profound capabilities in vision-language tasks but remain susceptible to object hallucination and visual misalignment. Investigations into this issue reveal that LMMs often misallocate attention to irrelevant visual tokens regardless of text input, known as "visual sink" tokens. These tokens cause the model to default to language priors rather than grounding its responses in the provided image. The Visual Attention Redistribution (VAR) model addresses this by dynamically identifying, suppressing and redistributing these visual sinks to "relevant" visual tokens, yet its standard implementation introduces a large computational costs, that mitigate its application.**

This report presents a rigorous architectural and empirical analysis of the VAR framework and introduces three simplified, algorithmically efficient variants designed to reduce computational latency while preserving visual grounding. By evaluating these models on multiple benchmarks, we demonstrate that lightweight approximations, such as Static Sink Biasing and KV-cache Sink Dropping, can achieve a optimal balance between inference latency and model accuracy. Our findings indicate that mitigating visual attention sinks significantly shifts probability distributions away from hallucinated tokens, offering a pathway toward robust, real-time multimodal inference.

I. Introduction

A. The Problem of Visual Hallucination

Large Multimodal Models (LMMs) have bridged the modality gap by combining pre-trained visual encoders to process image or video data with a large language model (LLMs) to generate a text output. Despite these advancements, LMMs suffer from a well-documented phenomenon known as object hallucination, in which the model generates plausible but factually incorrect text that contradicts the provided visual evidence. This issue stems from a variety of reasons including a disproportionate focus on a small subset of image tokens [4], over-reliance on parametric knowledge, statistical bias, biased feature learning, and attention deficiency toward images [1].

B. Visual Attention Sinks

Recent studies surrounding this phenomenon indicate these irrelevant visual tokens stems from the massive activation of specific dimensions in hidden states [2]. This extreme magnitude in a few specific dimensions is known as visual sink tokens, due to limited semantic information from the image [2]. These visual attention sinks act as "blind spots" that distract the model from semantic foreground objects, and are analogous to the attention sinks found in language models. Consequently, the model defaults to the statistical priors of its LLM backbone, answering questions based on training data biases rather than visual reality [3].

C. Motivation for VAR and Efficient Variants

To counteract this, mitigation strategies like Visual Attention Redistribution (VAR) actively intervene in the attention mechanism to suppress and redistribute these sink tokens. However, dynamic intervention at every layer and every head during autoregressive generation destroys highly optimized operations and introduces severe latency overhead. This report evaluates the theoretical foundation of the VAR framework and proposes three simplified, mathematically sound approximations to minimize computational overhead while retaining the anti-hallucination benefits of attention redistribution.

II. Related Work

The challenge of mitigating object hallucination in Large Multimodal Models (LMMs) has driven a rapid evolution of inference-time intervention strategies, transitioning from output-level logit manipulation to deep mechanistic attention routing. A foundational step in training-free mitigation was established through Visual Contrastive Decoding (VCD) [3]. VCD directly addresses the issue of LMMs over-reliance on statistical language priors, a primary driver of hallucination where models generate plausible text that contradicts the provided image. To solve this, VCD employs a contrastive inference method that generates two parallel sets of output logits: one conditioned on the original image, and another conditioned on a heavily distorted or noisy version of the image. By penalizing the logits produced by the distorted image, VCD effectively down-weights tokens generated purely from language biases and amplifies tokens that are strongly grounded in clear visual evidence. While highly effective, VCD intrinsically doubles the computational latency due to the necessity of paired forward passes for every generated token.

Subsequent methods sought to improve upon the performance and efficiency limitations of contrastive methods like VCD by investigating the structural ways LMMS perceive visual tokens. Woo et al. [4] systematically investigated the phenomenon of "blind spots" within large multimodal architectures. Their research demonstrated that LMMs do not process visual patches uniformly; rather, specific spatial and structural regions in the visual token sequence systematically fail to activate the LLM backbone, acting as informational voids that exacerbate hallucination rates. Building on the need for better text-to-vision grounding, An et al. [1] introduced Attention-Guided Alignment (AGLA). Moving beyond the post-hoc logit manipulation of VCD, AGLA targets the model's internal representations. By explicitly guiding cross-attention mechanisms, AGLA ensures a more robust semantic

alignment between the generated textual tokens and the corresponding visual features. Together, these works established that resolving hallucination requires addressing where the model is looking, rather than just filtering its final textual output.

The theoretical basis for manipulating internal attention distributions originates in pure Natural Language Processing NLP, Xiao et al. [5], who analyze the mechanics of autoregressive generation with their discovery of "attention sinks". Their work revealed that Transformer models inherently require a "dumping ground" to absorb excess attention probability mass to satisfy the softmax sum-to-one constraint. Consequently, LMMs arbitrarily allocate massive attention scores to initial sequence tokens (such as 'BOS' token), regardless of their semantic relevance. By exploiting this mechanistic quirk, Xiao et al. developed StreamingLLM, which permanently retains these attention sinks in the Key-Value (KV) cache, enabling LLMs to stream infinitely without architectural degradation. This work fundamentally proved that attention spikes do not always correlate with semantic importance, but are often artifacts of the softmax operation.

Building directly on this, Kang et al. (2025) extended the concept of attention sinks into the multimodal domain, providing the most direct foundational framework for our proposed methodology [2]. They identify that LMMs develop "Visual Attention Sinks" which accumulated massive attention weights in specific dimensions of hidden states. They act as the aforementioned "blind spots", starving the semantic foreground objects of attention and forcing the model to hallucinate based on language priors. To counter this, they proposed Visual Attention Redistribution (VAR), a training-free intervention that dynamically identifies these sink tokens, suppresses their weight, accumulates an attention budget, and then redistributes this weight to visual non-sink tokens. While proving powerful in reducing image hallucination, the VAR implementation requires layer-by-layer intervention that severely disrupts standard KV-cache optimizations and results in prohibitive computational increases. This architectural limitation forms the precise motivation for the simplified VAR variants proposed in this report, which seek to maintain the rigorous attention redistribution of the VAR framework while aggressively minimizing algorithmic complexity.

III. Methodology

The original Visual Attention Redistribution (VAR) framework operates on the premise that LMMs inherently generate uninformative "visual attention sinks"—tokens that accumulate disproportionate attention mass, thereby distracting the model from semantically relevant foreground objects. To mitigate this, the original VAR methodology employs a rigorous two-step dynamic intervention during the generation phase:

1. **Selecting Image-Centric Heads:** The model evaluates all attention heads across all layers to identify specific "image-centric" heads. This selection is based on measuring the concentration of visual attention sinks within those heads. Heads that disproportionately focus on these sink tokens are flagged for intervention.

2. **Redistributing Attention Budgets:** Within these specifically selected heads, VAR dynamically intercepts the softmax attention weights, It masks out the sink tokens, and redistributes their accumulated attention budget exclusively to the remaining, non-sink visual tokens.

While highly effective at restoring visual grounding, this token-level, head-specific redistribution requires intercepting the attention matrix at every generation step. Mathematically, this forces the model out of highly optimized C++ kernels (like FlashAttention) and into slower, memory-bound sequential processing, imposing an $O(N^2)$ computational overhead per generated token. To resolve this, we propose three simplified algorithms that approximate this redistribution using global masking, pre-calculated heuristics, and targeted hook wrappers.

A. Simplified Var-A: Static Sink Biasing

Empirical profiling of LLaVA-1.5-7B reveals that visual attention sinks are not entirely random; they frequently manifest at the extreme spatial boundaries (corners and edges) of the image patch sequence. VAR-A exploits this spatial predictability by utilizing a static heuristic, completely eliminating the need for dynamic sink discovery and recalculation.

Let the input sequence of length T contain a contiguous block of visual tokens $\mathcal{V}$ of length $N_{vis} = 576$. We identify the starting index of these visual tokens within the global sequence as p_{start}. The spatial boundary of the visual tokens spans the index range:

$$\mathcal{I}_{vis} = [p_{start}, p_{start} + N_{vis} - 1] \tag{1}$$

We define a fixed hyperparameter k representing the number of static sinks to drop from each end of the visual sequence (in our implementation, $k = 4$). The static sink indices, $\mathcal{S}_{static}$, are formulated by taking the union of the first k and last k indices of $\mathcal{I}_{vis}$:

$$\mathcal{S}_{static} = \{p_{start}, \ldots, p_{start} + k - 1\}$$
$$\cup$$
$$\{p_{start} + N_{vis} - k, \ldots, p_{start} + N_{vis} - 1\}$$

Prior to the initial prefill stage, the model's global 2D Boolean attention mask $M \in \{0, 1\}^{B \times T}$ (where batch size $B = 1$) is intercepted. The mask values corresponding to the static sink indices are zeroed out:

$$M_{0,i} = 0 \quad \forall i \in \mathcal{S}_{static} \tag{2}$$

By preemptively nullifying these static boundary patches at the global mask level, the self-attention mechanism is

mathematically forced to redistribute the attention probability mass ($\sum p_i = 1$) to the inner semantic tokens across all layers and heads. Because $\mathcal{S}_{static}$ is determined deterministically before the forward pass, this model introduces strictly $\mathcal{O}(1)$ computational overhead.

B. Simplified Var-B: Targeted Layer and Head Hooking

Variant B mimics the original VAR's dynamic adaptability by isolating interventions to highly image-centric feature spaces (specific layers and heads), but optimizes the latency by calculating the global sinks only once during a baseline discovery pass.

A standard forward pass is executed to extract the Key-Value (KV) cache. We isolate the Value cache $V^{(l)} \in \mathbb{R}^{B \times H \times T \times D}$ for each transformer layer $l \in \{1, \ldots, L\}$, where H is the number of attention heads and D is the head dimension. We slice the tensor to isolate only the visual tokens

$$V_{vis}^{(l)} = V_{0,\ldots,\mathcal{I}_{vis}}^{(l)}$$

For each layer l, we compute the L_2 norm along the head dimension D to measure the magnitude of the Value representations, yielding a matrix of norms. We then average these norms across all H attention heads to obtain a layer-specific sink score vector $\bar{n}^{(l)} \in \mathbb{R}^{N_{vis}}$:

$$\bar{n}_j^{(l)} = \frac{1}{H} \sum_{h=1}^{H} \left\| V_{0,h,j,:}^{(l)} \right\|_2 \quad \forall j \in \{1, \ldots, N_{vis}\} \tag{3}$$

To compute the global sink score for each visual token, we average $\bar{n}^{(l)}$ across all L layers:

$$S_j = \frac{1}{L} \sum_{l=1}^{L} \bar{n}_j^{(l)} \tag{4}$$

We identify the relative indices of the top K tokens (where $K = 50$) with the highest global sink scores, and map them back to their absolute positions in the global sequence:

$$\mathcal{S}_{global} = \{i + p_{start} \mid i \in \text{argtopK}(S)\} \tag{5}$$

We define a subset of target layers $\mathcal{L}_{target}$ (e.g., the final 4 layers) and target heads $\mathcal{H}_{target}$ (e.g., the first 50% of heads). A custom wrapper is injected into $\mathcal{L}_{target}$. During the autoregressive forward pass, the wrapper intercepts the 4D attention mask $M^{(4D)} \in \mathbb{R}^{B \times H \times Q \times K_{seq}}$. For the targeted heads and discovered sink indices, the mask is set to the minimum representable float ($-\infty$), ensuring the softmax attention weight evaluates to zero:

$$M_{0,h,q,k}^{(4D)} = \text{min_dtype} \quad \text{if } h \in \mathcal{H}_{target} \text{ and } k \in \mathcal{S}_{global} \tag{6}$$

This targeted penalty replicates the original VAR by constraining the redistribution of attention budgets strictly to the heads responsible for deep visual-textual alignment.

C. Simplified VAR-C: Global KV-Cache Sink Dropping (Aggressive Approximation)

Variant C hypothesizes that if a visual token acts as a severe mathematical sink across the network, it is universally uninformative and can be dropped globally, trading fine-grained head-specific control for massive computational speedups during decoding.

Variant C utilizes the exact same mathematical formulation as Variant B (Equations 4, 5, and 6) to compute the L_2 norm of the V-cache, aggregate the sink scores S, and extract the top $K = 50$ global sink indices, $\mathcal{S}_{global}$.

Instead of utilizing layer-specific wrappers that intercept the 4D attention mask during decoding, Variant C intervenes strictly at the input level. Prior to generating any new tokens, the original 2D global Boolean attention mask $M \in \{0, 1\}^{B \times T}$ is aggressively modified. The indices corresponding to the dynamic sinks are zeroed out:

$$M_{0,i} = 0 \quad \forall i \in \mathcal{S}_{global} \tag{7}$$

The model then executes a *second* prefill pass using this globally modified mask to generate the first token's logits. Let LLM_θ denote the transformer model. The dynamically regularized KV-cache is generated as:

$$\text{logits}_{t_1}, \text{KV}_{regularized} = \text{LLM}_\theta(\text{Input}, \text{AttentionMask} = M) \tag{8}$$

By forcing the entire transformer architecture to ignore the dynamic sinks universally during the prefill stage, the resulting $\text{KV}_{regularized}$ state is inherently grounded in the semantic foreground. Consequently, the subsequent autoregressive decoding loop requires zero Python-level intervention or mask manipulation. The model seamlessly utilizes standard, hardware-optimized generation protocols (e.g., FlashAttention), decoding at near-baseline speeds while retaining the image-specific adaptability of dynamic sink discovery.

D. Dataset Description

To evaluate these variants efficiently, we sample from three distinct multimodal benchmarks. These include:

- **MME:** A comprehensive general vision-language task benchmark measuring perception and cognition. These questions focus on (Yes/No) questions, where each image is given two questions labeled differently (one yes, one no). Scores are measured via Accuracy and Accuracy+ (which requires both questions in the image pair to be correct). The overall score is the sum of these accuracies scores. For computation efficiency and hardware limitations, a randomized subset of the full MME benchmark is used. The randomized subset includes 100 total images each with 2 (Yes/No) questions per image. [6]

- **POPE:** A benchmark designed to measure visual hallucination tasks, POPE is evaluated using Accuracy, Precision, Recall, and F1 score. A randomized subset

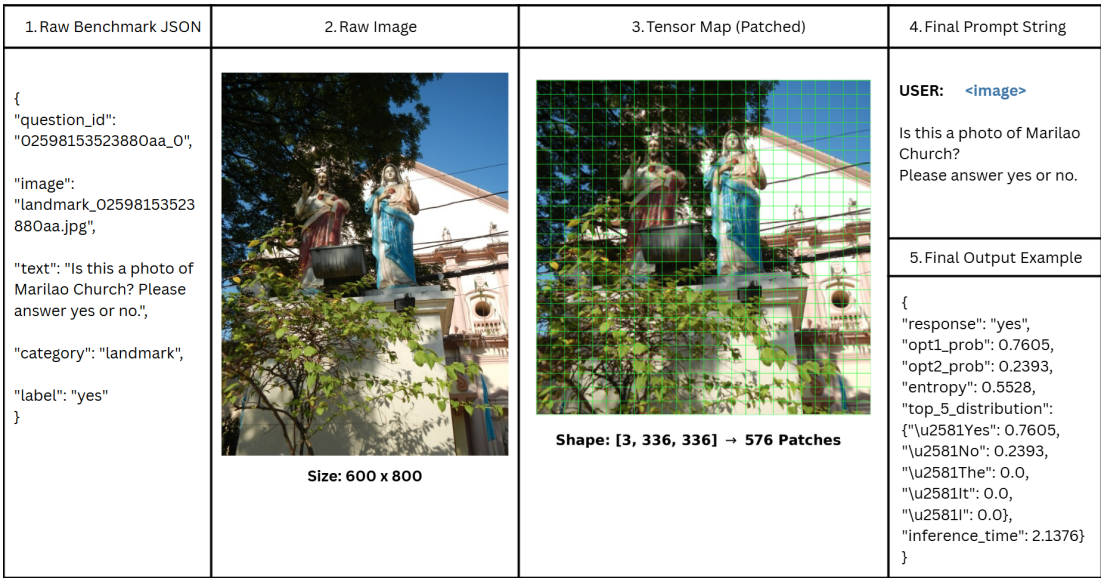

Fig. 1. Visual data preprocessing pipeline for LLaVA-1.5-7B. The raw input image (Marilao Church) is center-cropped and resized to a canonical 336×336 resolution. The 24×24 grid overlay illustrates the CLIP ViT-L/14 patching mechanism, which decomposes the image into 576 discrete 14×14 pixel patches. These 576 visual tokens constitute the sequence over which visual attention sinks dynamically form and are subsequently regularized.

is also used to evaluated this benchmark for the same reason as with the MME benchmark. The final subset includes 80 total questions, with 4 (Yes/No) questions per image (equally labeled). [7]

- **MMVP:** This benchmark focuses on vision-centric tasks, and is formatted using multiple-choice questions (a/b). An accuracy score is used to measure performance with the benchmark including a total of 300 images. [8]

Evaluation of each of the 3 benchmarks follows the LLaVA evaluation methodology [9]. This includes a pre-determined script that scores each of the model's output. For the MMVP benchmark, performance is scored using Gemini-3.1, as instructed by the MMVP repository [8].

IV. Implementation

A. Environment and Experimental Setup

To evaluate the computational and performance trade-offs of the original Visual Attention Regularization (VAR) framework and our proposed variants, we deployed a highly controlled, standardized inference environment.

All experiments were executed on a cloud-based compute instance equipped with a single NVIDIA L4 Tensor Core GPU (22.5 GB VRAM) running an Ubuntu 22.04 LTS operating system. This high-memory accelerator ensures that memory bandwidth limitations do not artificially inflate the inference latency metrics, providing a fair baseline for $O(1)$ versus $O(N^2)$ algorithmic comparisons.

The experimental pipeline was built utilizing Python 3.10 and the PyTorch (v2.1.0) deep learning framework. The Large Multimodal Model architecture was instantiated using the HuggingFace transformers library. To accommodate the massive parameter count of the baseline LLaVA-1.5-7B model while reserving sufficient VRAM for the extensive Key-Value (KV) caching required by VAR-B and VAR-C prefill passes, the model was loaded using 4-bit NormalFloat (NF4) quantization via the bitsandbytes library. Specifically, the base weights were quantized to 4-bit precision, but the computational datatype (bnb_4bit_compute_dtype) was explicitly set to torch.float16 to maintain gradient and activation precision during the forward passes.

To isolate the true algorithmic latency of each method, all models were evaluated under strict torch.inference_model() context managers, which disable gradient tracking and reduce memory overhead. Furthermore, high-precision temporal tracking (time.perf_counter()) was surgically injected to wrap only the autoregressive prefill and decoding blocks. This explicitly excludes the overhead of disk I/O, image loading, and initial tokenization, ensuring that the reported "Inference Time" metric purely reflects the architectural cost of the attention masking and hook interventions.

B. Data Preprocessing, Splitting, and Evaluation Approach

Due to the profound computational latency of the original VAR model (which breaks hardware-level FlashAttention protocols in favor of manual, step-by-step sequential decoding loops), evaluating the entirety of the MME, POPE, and MMVP benchmarks was computationally prohibitive. To maintain statistical significance while ensuring feasibility, we employed a randomized subset as explained previously.

The evaluation required strict homogenization of both visual and textual modalities before ingestion by the LLaVA-1.5-7B baseline.

TABLE I
MME BENCHMARK RESULTS

Model	Accuracy (%)	Accuracy+ (%)	Total Score	Avg Target Prob	Avg Entropy	Avg Leakage (%)	Avg Inf Time (s)
Baseline	72.50	47.0	119.5	0.6762	0.4809	0.02	0.3808
VAR	75.50	52.0	127.5	0.6936	0.4414	0.03	1.5691
Variant A	74.00	51.0	125.0	0.6773	0.4820	0.02	0.3950
Variant B	73.00	48.0	121.0	0.6762	0.4810	0.02	0.7124
Variant C	73.00	47.0	120.0	0.6738	0.4856	0.02	0.7155

TABLE II
POPE BENCHMARK RESULTS

Model	Precision (%)	Recall (%)	Avg Target Prob	Avg Entropy	Avg Leakage (%)	Avg Inf Time (s)
Baseline	85.0	85.0	0.8173	0.3368	1.37	1.8751
VAR	88.64	97.5	0.8312	0.3477	1.53	9.9127
Variant A	82.93	85.0	0.8149	0.3441	1.40	1.9348
Variant B	85.0	85.0	0.8173	0.3368	1.37	2.1557
Variant C	85.0	85.0	0.8142	0.3547	1.44	2.2566

TABLE III
MMVP BENCHMARK RESULTS

Model	Accuracy (%)	Avg Target Prob	Avg Entropy	Avg Leakage (%)	Avg Inf Time (s)
Baseline	58.00	0.4141	0.9868	31.08	0.7643
VAR	61.33	0.3092	1.0285	45.24	3.2497
Variant A	59.67	0.3911	1.0059	32.96	0.8488
Variant B	58.00	0.4141	0.9868	31.07	1.0855
Variant C	60.00	0.3890	1.0049	33.45	1.1669

1) **Visual Preprocessing:** Raw RGB images were passed through the model's integrated CLIP ViT-L/14 image processor. This resizes the images to the canonical 336×336 resolution, normalizes the pixel values utilizing standard ImageNet mean and standard deviation matrices, and converts them into 576 discrete patch embeddings (24×24 grid).
2) **Textual Prompt Formatting:** To properly trigger the model's instruction-following capabilities, raw benchmark questions were concatenated into the exact conversational template utilized during LLaVA's instruction fine-tuning phase.

A visual representation of this process is displayed in figure 1. The raw benchmark json displays the initial text input of a question, in which the related image is linked to the text prompt. The raw image is cropped and resized to a 336×336 resolution, decomposing the image into the necessary 14×14 grid to process the information. The model then receives these 576 visual tokens, the raw text tokens, and the system as inputs. After running the model, it produces a text input similar to that displayed in the final output example, which includes the response, the output probabilities, the entropy, a distribution of the top 5 most probably text tokens, and the inference time.

C. Dynamic Evaluation and Target Probability Extraction

Unlike standard benchmark evaluations that rely strictly on final string matching (which provides a binary 1 or 0 accuracy score), our approach extracts deep, continuous metrics from the model's internal logit distributions to mathematically quantify "hallucination" and "confidence."

During the decoding of the first generated token (the decision token), the algorithm extracts the raw, un-normalized logits $z \in R^V$, where V is the vocabulary size. To mitigate PyTorch FP16 underflow errors - which frequently result in NaN values during operations on highly skewed distributions - the logits are explicitly cast to float32 before applying the Softmax function:

$$\mathbf{p} = \text{Softmax}(\mathbf{z}_{FP32})$$

Using dynamic token ID resolution via the processor's tokenizer, we aggregate the probability mass for the expected answers (e.g., summing the probabilities of ["Yes", " Yes", "yes", " yes"] for binary tasks, or ["(a)", " a", "A"] for MMVP). This yields:

1) **Target Probability** (p_{target})**:** The model's raw confidence in the ground-truth answers.
2) **Entropy (H):** Calculated as $H = -\sum_i p_i log(p_i + \epsilon)$. This quantifies the internal chaos or uncertainty of the model; lower entropy indicates sharp visual grounding, while high entropy indicates confusion and susceptibility to hallucination.
3) **Probability Leakage:** Defined as $1 - (p_{opt1} + p_{opt2})$. This measures the probability mass erroneously assigned to hallucinated starting tokens (e.g., "The", "Based", "I") rather than the explicitly requested instruction formats.

By utilizing these continuous evaluation metrics, we can assess whether the simplified VAR algorithms successfully force the model to ground its reasoning in the visual features, long before the final textual output is generated.

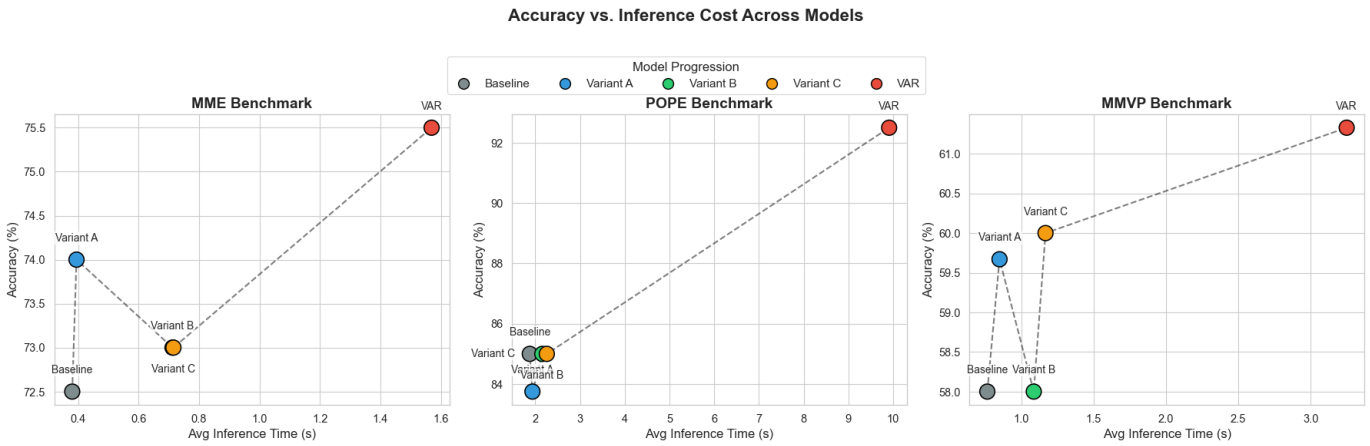

Fig. 2. Scatterplot displaying the average inference time (s) for each model against its Accuracy (%) among each benchmark. The VAR model naturally has the highest accuracy in all 3 benchmarks but suffers from roughly 4-5 fold increases in average inference time (s)

V. Results and Discussion

Consistent with the findings of Kang et al. (2025), the original Visual Attention Regularization (VAR) framework achieved the highest overall performance across all benchmarks. On MME, VAR improved the Total Score to 127.5 (baseline: 119.5); on POPE, it achieved a remarkable 97.5% Recall and 88.64% Precision; and on MMVP, it elevated accuracy to 61.33% (baseline: 58.00%).

However, this superior visual grounding comes at a severe computational cost. Because VAR mandates dynamic, layer-by-layer attention mask interception, it entirely bypasses hardware-accelerated KV-cache optimizations. Consequently, VAR exhibits profound latency degradation, requiring 1.56s per generation on MME (a $4.1\times$ increase over the baseline's 0.38s) and an astonishing 9.91s on POPE (a $5.2\times$ increase). While optimal for performance, this high inference time makes VAR unpractical in some real-time streaming or high-throughput applications, directly motivating the necessity of our simplified variants.

A. Variant A (Static Sink Biasing)

Variant A, which completely eliminates dynamic discovery in favor of masking static spatial boundaries, emerged as the most computationally efficient approximation. As expected, it introduced virtually zero overhead, matching the baseline inference times (e.g., 0.39s on MME, 1.93s on POPE).

Surprisingly, on general perception tasks, this naive heuristic captured a vast majority of the original VAR's benefits. On MME, Variant A achieved a Total Score of 125.0, aggressively closing the gap toward the original VAR (127.5). On MMVP, it achieved 59.67% accuracy. This confirms our hypothesis that visual attention sinks frequently accumulate at the physical boundaries of the image patch sequence; simply blinding the model to these perimeters naturally forces attention toward the semantic center.

However, Variant A performed unexpectedly worse than the baseline on the POPE benchmark, with Precision dropping to 82.93% (baseline: 85.0%). POPE explicitly queries the existence of specific, often peripheral, objects. We hypothesize that statically masking the outermost image patches may inadvertently delete critical visual evidence if the queried object resides at the edge of the frame, leading to increased false positives or false negatives. While highly effective for centralized visual reasoning (MME/MMVP), Variant A's spatial rigidity proves detrimental for exhaustive spatial polling. Further studies utilizing bounding-box analysis are required to conclusively map Variant A's failure modes on peripheral objects.

B. Variant B (Targeted Hooking)

Variant B attempted to balance adaptability and speed by discovering sinks dynamically but restricting interventions to a subset of deep layers and heads. This yielded moderate success on MME (Total Score: 121.0), but yielded results identical to the baseline's Accuracy and Precision on both POPE and MMVP, while still suffering a moderate time penalty (e.g., 2.15s on POPE vs baseline 1.87s).

This lack of deviation from the baseline on complex tasks was unexpected. The results suggest that partial, targeted masking during the decoding phase is insufficient to overcome the LLM's deeply ingrained language priors. Because the unhooked heads and earlier layers are still permitted to attend to the visual sinks, they likely accumulate enough statistical mass to overpower the regularized heads during the final softmax token selection. Consequently, the model defaults back to the baseline's hallucinatory trajectory.

C. Variant C (KV-Cache Sink Dropping)

Variant C leverages a two-pass prefill to globally drop dynamic sinks from the KV-cache, ensuring standard decoding speeds thereafter. This method demonstrated notable success on the MMVP benchmark, achieving 60.00% accuracy (surpassing Variant A and approaching original VAR's 61.33%).

MMVP tests fine-grained visual details (e.g., text, textures, orientation) where models typically struggle to differenti-

ate between subtle (a)/(b) choices. Variant C's success here aligns with theoretical expectations: by dynamically identifying mathematical sinks specific to the complex image and globally purging them from the entire transformer architecture during prefill, the resulting KV-cache is fundamentally grounded in semantic details. While it incurs a moderate prefill time penalty (1.16s on MMVP vs baseline 0.76s), it provides a robust, adaptable solution that avoids the peripheral-blindness flaw of Variant A.

D. Anomalies in Internal Probability Metrics

Analysis of the raw probability distributions revealed an unexpected anomaly regarding Probability Leakage (the probability mass assigned to non-instructional tokens). On the MMVP benchmark, the baseline model exhibited a leakage of 31.08%. Counter-intuitively, the original VAR model—despite achieving the highest accuracy—saw its leakage spike to 45.24%.

Typically, higher leakage indicates a loss of instruction-following capability (i.e., the model begins outputting conversational filler instead of the requested "(a)" or "(b)"). The fact that VAR is highly accurate but suffers from massive probability leakage suggests that severe, localized manipulation of the attention matrix may disrupt the structural formatting priors of the LLM. Essentially, the model sees the correct visual answer but struggles to format it immediately. This points to a latent architectural fragility when intervening in LLM attention mechanisms, emphasizing that further research is required to understand how visual sink suppression mathematically interacts with instruction-tuning alignments.

VI. Conclusion and Future Work

Large Multimodal Models (LMMs) possess profound capabilities but are fundamentally constrained by their susceptibility to object hallucination. This phenomenon is heavily exacerbated by "visual attention sinks"—uninformative spatial patches that erroneously accumulate attention mass, blinding the model to semantic foreground objects and forcing a reliance on statistical language priors. While the original Visual Attention Redistribution (VAR) framework effectively suppresses these sinks to restore visual grounding, its requirement for dynamic, layer-by-layer mask interception introduces a severe computational bottleneck that precludes real-time deployment.

In this report, we successfully designed, implemented, and evaluated three mathematically simplified variants of the VAR framework to address this latency-performance dichotomy. By utilizing spatial heuristics and prefill-stage interventions, our models demonstrated that algorithmic approximations can successfully redistribute attention budgets without breaking hardware-level decoding optimizations. Most notably, Variant A (Static Sink Biasing) achieved a Total Score of 125.0 on the MME benchmark—aggressively closing the gap toward the computationally exhaustive original VAR (127.5)—while maintaining an average inference time of 0.3950 seconds, virtually identical to the unregularized baseline. Ultimately, this research validates that lightweight, training-free architectural interventions can mitigating hallucinations while preserving the high-throughput inference necessary for practical LMM deployment.

A. Future Work

While the proposed simplified VAR models demonstrate strong potential, the varied outcomes across different visual tasks illuminate several critical avenues for further study:

- **Expansion to Broader Benchmarks:** The performance degradation of Variant A on the POPE benchmark suggests that static spatial masking may inadvertently delete peripheral objects. Future research must evaluate these variants across the full, un-subsetted versions of MME, POPE, and MMVP, as well as highly adversarial datasets like the Object Hallucination Benchmark (OHBench) and traditional VQA-v2. Incorporating bounding-box spatial analysis will help definitively map the spatial failure modes of heuristic masking.
- **Scaling to Larger Parameter Models:** The current experiments were constrained to the 7B parameter regime. Further studies should test the scalability of our dynamic prefill method (Variant C) on significantly larger models (e.g., 13B, 34B, and 70B parameter variants). Investigating whether the density and magnitude of visual attention sinks scale linearly with LLM capacity will provide deeper insights into the mechanistic origins of these blind spots.
- **Evaluation Across Diverse Multimodal Architectures:** LLaVA-1.5 relies on a simple MLP projection matrix to bridge the Vision Transformer (ViT) and the LLM. It remains an open question whether visual attention sinks are a unique artifact of this specific bridging paradigm. Future testing must apply these VAR variants to fundamentally different architectures, such as models utilizing Q-Former cross-attention mechanisms (e.g., BLIP-2, InstructBLIP) or natively integrated multimodal foundations (e.g., Llama-3-Vision, Qwen-VL), to determine the universality of the visual sink phenomenon.
- **Investigating Instruction-Tuning Degradation:** As observed in the MMVP probability leakage metrics, aggressively altering the attention matrix can disrupt the model's structural formatting priors, causing it to "forget" how to output the requested response format (e.g., generating conversational filler instead of "(a)" or "(b)"). Future mechanistic interpretability studies should investigate how visual sink suppression mathematically interacts with instruction-tuned alignment layers, aiming to decouple visual grounding interventions from linguistic formatting degradation.

References

[1] Wenbin An, Feng Tian, Sicong Leng, Jiahao Nie, Haonan Lin, QianYing Wang, Guang Dai, Ping Chen, and Shijian Lu. (2024). AGLA: Mitigating object hallucinations in large vision-language models with assembly of global and local attention. arXiv. https://arxiv.org/html/2406.12718v2.

[2] Seil Kang, Jinyeong Kim, Junhyeok Kim, Seong Jae Hwang. (2025). See What You Are Told: Visual Attention Sink in Large Multimodal Models. arXiv. https://arxiv.org/abs/2503.03321.
[3] Sicong Leng, Hang Zhang, Guanzheng Chen, Xin Li, Shijian Lu, Chunyan Miao, Lidong Bing. (2023). Mitigating object hallucinations in large vision-language models through visual contrastive decoding. arXiv. https://arxiv.org/abs/2311.16922
[4] Sangmin Woo, Donguk Kim, Jaehyuk Jang, Yubin Choi, Changick Kim. (2025). Investigating blind spots in large multimodal models. arXiv. https://arxiv.org/abs/2405.17820
[5] Guangxuan Xiao, Yuandong Tian, Beidi Chen, Song Han, Mike Lewis. (2023). Efficient streaming language models with attention sinks. arXiv. https://arxiv.org/abs/2309.17453.
[6] Fu, Chaoyou and Chen, Peixian and Shen, Yunhang and Qin, Yulei and Zhang, Mengdan and Lin, Xu and Yang, Jinrui and Zheng, Xiawu and Li, Ke and Sun, Xing and others. (2023). MME: A Comprehensive Evaluation Benchmark for Multimodal Large Language Models. arXiv. https://arxiv.org/abs/2306.13394
[7] Yifan Li, Yifan Du, Kun Zhou, Jinpeng Wang, Wayne Xin Zhao, Ji-Rong Wen. (2023). Evaluating Object Hallucination in Large Vision-Language Models. arXiv. https://arxiv.org/abs/2305.10355
[8] Shengbang Tong and Zhuang Liu and Yuexiang Zhai and Yi Ma and Yann LeCun and Saining Xie. (2024). Eyes Wide Shut? Exploring the Visual Shortcomings of Multimodal LLMs. arXiv. https://arxiv.org/abs/2401.06209.
[9] Haotian Liu, Chunyuan Li, Qingyang Wu, Yong Jae Lee. (2023). Visual Instruction Tuning. arXiv. https://arxiv.org/abs/2304.08485